

6 Electrochemistry

AS · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Oxidation is electron loss, reduction electron gain. Oxidation numbers help balance redox equations, but charge and atoms must also balance. Standard electrode potentials describe reduction half-cells under standard conditions; positive $E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$ indicates thermodynamic feasibility under standard conditions, not reaction rate.

Worked example: follow the reasoning

For Zn^{2+}/Zn -0.76 V and Cu^{2+}/Cu $+0.34$ V, Zn oxidises and $E^\circ_{\text{cell}} = 0.34 - (-0.76) = +1.10$ V.

Quick check · multiple choice

A Which statement identifies oxidation?

[1]

- A Gain of protons
- B Loss of electrons
- C Loss of neutrons
- D Gain of electrons

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Given $E^\circ(\text{Fe}^{3+}/\text{Fe}^{2+}) = +0.77$ V and $E^\circ(\text{I}_2/\text{I}^-) = +0.54$ V, find E°_{cell} and identify the species oxidised.

[3]

Final answer: _____

Answers and explanation

Quick check: B. Loss of electrons

Question 1 [3 marks]

Fe^{3+} reduced; I^- oxidised [1]. $E^\circ_{\text{cell}} = 0.77 - 0.54 = +0.23 \text{ V}$ [2].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Chemistry 9701, 2025–2027. <https://www.cambridgeinternational.org/Images/664563-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.